NGEE ANN POLYTECHNIC
SCHOOL OF DESIGN & ENVIRONMENT
DIPLOMA IN ENGINEERING (BUILDING SERVICES & FIRE SAFETY)

Module : **Air-conditioning & Mechanical Ventilation**

Experiment No : ACMV – Practical 1

Title of Experiment : Performance of a Water-cooled Chiller and Pumps of an Air-conditioning Plant

Objectives :
1. To determine the chiller cooling capacity, rate of heat rejection and efficiency.
2. To determine the chilled and condenser water pump efficiencies.
3. To determine the air-conditioning system efficiency.
4. To study the construction of the chiller.

Equipment : Air-conditioning System serving the Integrated Facilities Management Lab

Theory:

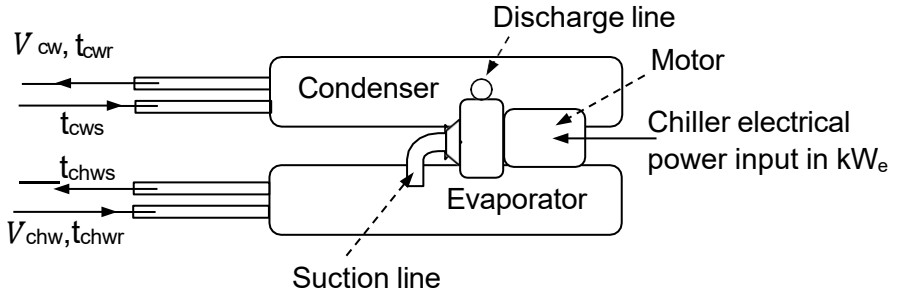


Figure 1 – Water-cooled Chiller

1. Chiller Cooling Capacity

Chiller cooling capacity in kW = $\dot{m}_{chw}$. c_w ($t_{chwr} - t_{chws}$) kW.....(1)

Chiller cooling capacity in Refrigeration Ton (RT) = $\frac{\dot{m}_{chw} . c_w (t_{chwr} - t_{chws})}{3.517}$ (RT) ..(2)

Where $\dot{m}_{chw}$ = mass flow rate of chilled water, kg/s
c_w = specific heat capacity of water = 4.2 kJ/kgK
t_{chws} = chilled water supply temperature, °C
t_{chwr} = chilled water return temperature, °C

2. Chiller Heat Rejection Rate

Chiller heat rejection rate = $\dot{m}_{cw}$.c_w ($t_{cwr} - t_{cws}$) kW.....(3)

Where $\dot{m}_{cw}$ = mass flow rate of condenser water, kg/s
c_w = specific heat capacity of water = 4.2 kJ/kgK
t_{cws} = condenser water supply temperature, °C
t_{cwr} = condenser water return temperature, °C

Chiller heat rejection rate can also be determined by:

Chiller heat rejection rate in kW

= chiller cooling capacity (kW) + chiller electrical input power (kW_e) ..... (4)

3. Chiller Efficiency

Chiller efficiency is sometimes expressed in kW/RT and can be calculated as follows:

$$\text{Chiller efficiency} = \frac{\text{Chiller Electrical Input Power}}{\text{Chiller Cooling Capacity in RT}} \quad \text{kW/RT} \quad \ldots\ldots (5)$$

4. Pump Output Power

Pump output power (kW)

$$= \frac{\text{Volume flow rate in l/s}}{1000} \times (\text{Outlet pressure} - \text{Inlet pressure}) \text{ in kPa} \quad \ldots\ldots (6)$$

Pump input power = Motor Output Power
= Motor Input Power x motor efficiency ..... (7)

(Note : Assume a motor efficiency of 85%)

$$\text{Pump Efficiency} = \frac{\text{Pump Output Power}}{\text{Pump Input Power}} \times 100\% \quad \ldots\ldots (8)$$

5. Ratio of pump motor input power to water flow rate

$$= \frac{\text{Pump Motor Input Power}}{\text{Water volume flow rate}} \quad \text{kW/m}^3\text{/s}$$

For condenser water pump:

$$= \frac{\text{Condenser Water Pump Motor Input Power .}}{\text{Condenser Water Volume Flow Rate}} \quad \text{kW/m}^3\text{/s} \quad \ldots\ldots (9)$$

For chilled water pump:

$$= \frac{\text{Chilled Water Pump Motor Input Power .}}{\text{Chilled Water Volume Flow Rate}} \quad \text{kW/m}^3\text{/s} \quad \ldots\ldots (10)$$

6. System Efficiency

$$\text{System Efficiency} = \frac{\sum \text{Power Consumption by Chillers, Pumps \& Cooling Towers}}{\text{Cooling Capacity in RT}} \quad \text{kW}_e\text{/RT}$$

…… (11)

Procedure

1. Check that the chiller plant has been in operation for at least 20 mins and the chiller is still running.

2 At the monitor of the BMS system, click on the Chiller Plant schematic in the left panel to display the plant graphics with readings and dashboard.

At a particular instant in time, take an image (Image 1) of the Chiller Plant graphics with readings. Immediately click on the Parameter button and take an image (Image 2) of the readings on the pumps and chillers. Click on the Power Meter graphics in the left panel and take an image (Image 3) of the power meter readings.

3. From Image 3, read the Chiller Electrical Input Power, Chilled Water Pump Motor Input Power, Condenser Water Pump Motor Input Power and Cooling Tower Motor Input Power and record in Table 2.

4. From Image 2, read the suction and discharge pressures of the operating chilled water pump and condenser water pump and record in Table 3.

5. Study the chiller and complete Table 4 based on your observations and measurements.

Results and Calculations

1. Calculate the chiller operating cooling capacity in kW and RT using Equations (1) and (2) respectively. Record in Table 5

2. Calculate the chiller heat rejection rate using Equation (3). Record in Table 5.

3. Calculate the chiller heat rejection rate using Equation (4). Record in Table 5.

4. Calculate the chiller efficiency in input kW_e/RT using Equation (5). Record in Table 5.

5. Compute the chilled and condenser water pump input and output powers, efficiencies and ratio of pump motor input power to water flow rate for the condenser and chilled water circuits using Equations 6 to 10. Record in Table 6.

6. Calculate the system efficiency using Eqn. 11. Record in Table 7.

7. Draw and/or label the main components of the chiller, pipework, accessories, sensors and fittings onto Figure 1. Use the symbols given in the legend wherever applicable.

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Module : **Air-conditioning & Mechanical Ventilation**

Experiment No : ACMV P3 Total Points: **/30**

Title of Experiment : Performance of a Water-cooled Chiller and Pumps of an Air-conditioning Plant

Tutorial Grp: **A / B***

Name: _____ **Lab Grp:** **1 / 2 / 3 / 4***

Date of Experiment: 23 / 24* Jul 2025 **Submission Deadline:** **31 Jul 2025**

*Circle accordingly **Date Submitted:** _____

Results

Table 1

Equipment	Efficiency (kW_e/RT)
Cooling Tower	
Condenser Water Pump	
Chilled Water Pump	
Chiller	
System Efficiency	

(/1 point)

Table 2

	Parameter	Reading	
1	Chiller Electrical Input Power	kW	
2	Chilled Water Pump Motor Input Power	kW	
3	Condenser Water Pump Motor Input Power	kW	
4	Cooling Tower Motor Input Power	kW	
	Total Power Consumption	**kW**	
5	Chilled Water Supply Temperature	°C	
6	Chilled Water Return Temperature	°C	
7	Condenser Water Supply Temperature	°C	
8	Condenser Water Return Temperature	°C	
9	Chilled Water Flow Rate	l/s	kg/s
10	Condenser Water Flow Rate	l/s	kg/s

(/2 points)

Table 3

Equipment	From Pressure Gauges			
	Discharge Pressure		Suction Pressure	
Chilled Water Pump	bar*	kPa	bar*	kPa
Condenser Water Pump	bar*	kPa	bar*	kPa

***Conversion Factor: 1bar = 100 kPa**
For vacuum pressure, if applicable: 1 mm Hg = 100/760 kPa. (/1 point)

Table 4

Brand and model	
Type of refrigerant used in chiller	
Type of compressor	
Type of condenser	
Type of evaporator	
Type of metering device	

(/2 points)

Table 5

All steps of the calculations to be shown

Chiller cooling capacity in kW using Eqn 1	
Chiller cooling capacity in RT using Eqn 2	
Chiller heat rejection rate using Eqn 3	
Chiller heat rejection rate using Eqn 4	
Chiller efficiency in kW/RT using Eqn 5	

(/5 points)

Table 6

All steps of the calculations to be shown

Equipment	Pump Output Power (kW) (Eqn 6)	Pump Input Power (kW) (Eqn 7)	Pump Efficiency (%) (Eqn 8)	Ratio of Pump Motor Input Power to Water Flow Rate ($kW/m^3/s$)
Chilled Water Pump 1 or 2				**(Eqn 10)**
Condenser Water Pump 1 or 2				**(Eqn 9)**

(/4 points)

Table 7

All steps of the calculations to be shown

System Efficiency using Eqn 11	

(/1 point)

Legend

	Gate valve		Balancing valve	T	Thermometer
	Globe valve		2-way modulating valve	F	Flow meter
	Check valve		Flexible connection	PS	Pressure sensor
	Strainer	P	Pressure gauge	TS	Temperature sensor

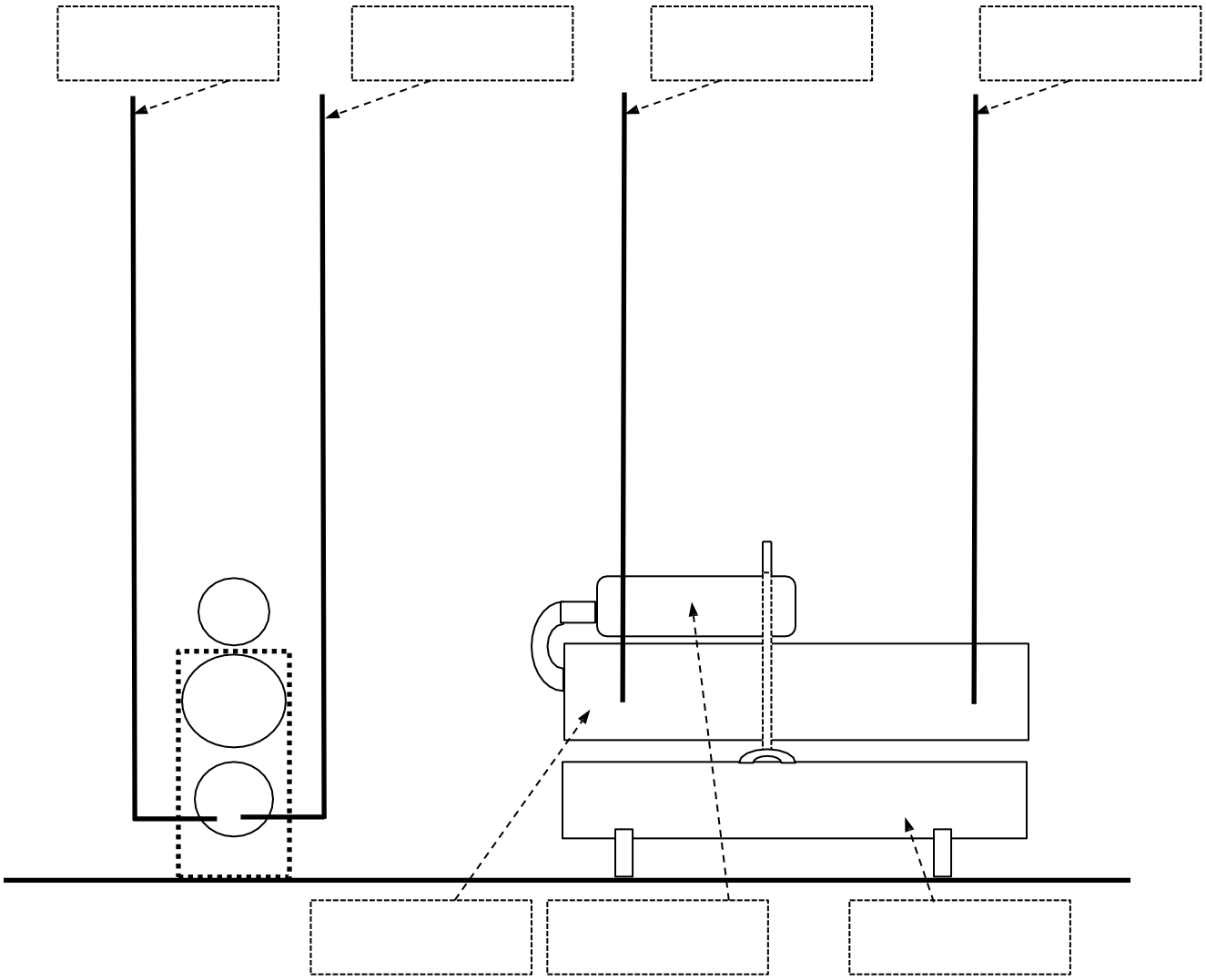


Figure 1 – End and Side Views of Chiller

(/6 points)

Comments

1. Chiller Efficiency.

   Compare the chiller kW_e/RT you have obtained with the normal range of values for such chillers (You would need to do some research on this.). If there are differences, discuss the possible causes?

(/4 points)

2. Ratio of Chilled Water Pump Motor Input Power to Chilled Water Flow Rate and Ratio of Condenser Water Pump Motor Input Power to Condenser Water Flow Rate

   Compare the values of the ratios you have obtained with the power limitations stipulated in SS 553:2016. i.e. 349 kW/m^3/s for chilled water system and 301 kW/m^3/s. Suggest the ways by which the efficiency of both systems can be improved.

(/2 points)

3. Comment on the calculated kW/RT of the chiller and system with that recorded from the BMS monitor. What are the possible reasons for the differences, if any?

(/2 points)